

LLMs for Multi-Agent System Control Project Proposal

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1 Motivation

Multi-agent systems (MAS) have vast potential for various applications of robotics. From search and rescue [1], to warehouse logistics [2] MAS can be used to improve efficiency and effectiveness in various domains. However, since the deployment of MAS is often expensive and complex, most MAS research is completed in simulation.

A significant body of research into MAS has been conducted in real-time strategy (RTS) games (e.g., StarCraft) [3]. In this medium, the game’s engine provides a clear set of actions for the agent and a coherent set of environmental constraints, while completion of the game’s objectives provides reinforcement learning algorithms a straightforward source of feedback on the performance of the agents [4]. The field has seen significant success in training agents for these games, so subsequent research seeks to apply these techniques to increasingly realistic and complex scenarios [5]. However, training agents in games is by no means a trivial task, and many real-world deployments still require a human in the loop to ensure that the agents are performing as expected. This is especially true in scenarios where the agents are operating in a dynamic environment with changing constraints and objectives.

With a human in the loop, an instinctive form of interface is natural language, therefore there has been a growing interest in using large language models (LLMs) to control agents in MAS. LLMs can be quite flexible in this regard. In the existing body of research, LLM control has been attempted in many levels of abstraction, including high level task allocation, mid level motion planning, and low level action generation [6]. For this project, we will be focusing on the high level task allocation problem, which has seen significant success. LLMs have been shown to generate high-quality task allocation plans for MAS in a variety of standard MAS and swarm problem formulations, including flocking and formation control [7], [8], [9] as well as robot soccer [10]. Although many different approaches have been attempted for LLM control of MAS, a particularly interesting approach involves having LLMs generate plans using a sort of intermediate representation. In [11], the authors have the LLM generate a behavior tree for the agents to follow. In [12], the authors use Linear Temporal Logic (LTL) as an intermediate representation for the LLM to generate plans.

While LLMs have been shown to be able to generate plans for MAS, there has been scant research conducted in evaluating these systems with a human in the loop and in situations with many robots. First, much of the previous work has focused on either pre-defined and common MAS problems (such as flocking and formation control) [9] or on a small number of robots without a human in the loop [7], [8], [13]. Second, relatively few studies have evaluated the quality of the user experience with these systems. In [12], the authors conducted a 4 participant user study to evaluate the usability of their system. However, this study was limited to the performance of the system and did not evaluate the user experience of the participants.

In this project, we will evaluate the effects that interface modality has on the performance and user experience of participants. We will compare these effects across a more traditional RTS game interface and a natural language interface. This work may inform more general design principles for MAS control interfaces for particularly complex scenarios.

2 Proposed Methodology

We will design a traditional RTS game interface and an LLM interface for controlling a MAS in a game environment. We will then perform a **within-subjects** user study to evaluate the performance and user experience of the two interfaces, with the ordering of the interfaces **counterbalanced** for each participant. A within-subjects design serves the function of controlling for the **confounding variable** of RTS game experience. Moreover, by counterbalancing the order of the interfaces' presentation to the participant, we seek to abate any **learning** or **sequence effects**.

We will evaluate a participant's performance with the MAS interface in terms of the score they achieve in the game environment. We will evaluate a participant's user-experience with the MAS interface in terms of task load and usability. To measure task load, we will be using the **NASA Task Load Index (NASA-TLX)** [14]. To measure usability, we will be using the **System Usability Scale (SUS)** [15]. Additionally, we will be collecting qualitative feedback from the participants in the form of a **Semi-Structured Interview** to gain insight into their experience with the two interfaces. Our **research hypothesis** is that the LLM interface will lead to a decreased task load and no meaningful difference in performance when compared to the RTS game interface.

The first major part of our project is creating a game environment for the MAS to operate in. We will be using the Robotarium [16] as our MAS platform and we will utilize the Robotarium's overhead projector to create a game environment for the MAS to operate within. While we have not yet decided specific elements of the game environment, it should satisfy the following:

- challenging enough to induce task saturation at times
- require the simultaneous coordination of a large number of robots
- include both positive rewards to serve as goals and negative rewards to serve as constraints

Much of the previous work has focused on contrived examples that do not require conforming to a set of environmental constraints, whereas these are incredibly important aspects of real-world MAS. We will be creating a scoring system for the game that will provide positive feedback to the MAS for completing tasks and negative feedback for failing to complete tasks or violating constraints.

With the Robotarium providing a software simulator, hardware, and runtime environment for the MAS, we will be able to focus on creating the game environment and interfaces for controlling the MAS. The team has not yet decided on the specific LLM for use in the project, but selecting the LLM will be one of the first tasks for the project. Additionally, we will be creating a traditional RTS game interface for controlling the MAS. A technology stack has also not yet been selected for the RTS game interface, but we intend for the design to adhere to the human-computer interaction principles for RTS games as presented in [17] and [18].

3 Proposed Evaluation

The project is considered successful if the following criteria are met:

- A game environment is created utilizing the Robotarium platform
- A real-time strategy game interface is created for controlling the MAS
- An LLM interface is created for controlling the MAS
- A user study is conducted to evaluate the performance and user experience of the two interfaces

4 Project Timeline

4.1 Project Check-In Day 1

By project check-in day 1, the following milestones should ideally be completed:

- The game environment should be established and built within the Robotarium platform.
- The real-time strategy game interface should be designed and partially implemented

- The LLM interface should be designed and partially implemented

4.2 Project Mid-Point "Poster" Session

By the project mid-point poster session, the following milestones should ideally be completed (in addition to the milestones above):

- The real-time strategy game interface should be fully implemented and functional
- The LLM interface should be fully implemented and functional
- The user study should be designed and ready to be conducted
- The mid-point poster should be completed and presented

4.3 Project Check-In Day 2

By project check-in day 2, the following milestones should ideally be completed (in addition to the milestones above):

- The user study should be conducted and completed

4.4 Final Project Presentations

By the Final project presentation, the following milestones should ideally be completed (in addition to all other milestones above):

- The user study should be analyzed and the results should be presented
- The final project poster should be completed and presented

5 Proposed Grading Rubric

Proposed Grading Rubric	
Finding / System Component / Capability / Validation Test	Points
Implementation of a Complex Game on the Robotarium	30
Intuitive, Consistent, and Accessible RTS Interface	30
Intuitive, Consistent, and Accessible LLM Interface	30
Setup and Conduct User Study	30
User Study Data Analysis and Interpretation	30

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